

Colleen Milos

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EDUCATION

University of Illinois, Grainger College of Engineering

Expected May 2028

Bachelor of Science in Mechanical Engineering & Innovation, Leadership, and Engineering Entrepreneurship

Relevant Coursework: Heat Transfer, Design for Manufacturability, Mechanical Design, Intro to Computer Science (Java) (C++)

SKILLS

Technical: CREO, NX, Siemens Teamcenter, Creo Simulate, Fusion 360, Cura, C++, Java, HTML, Python, aPriori, Procore, Bidder Editor

Certifications & Training: SAE International GD&T, Xcelerator Academy NX, PTC Creo University

Manufacturing: TIG Welding, Waterjet, Lathe, Mill

PROFESSIONAL EXPERIENCE

Caterpillar Inc.

Peoria, Illinois

Parallel Co-op Engineering

May 2025 - present

- Designing a 3-box countershaft transmission from its inception with Excel and Creo.
- Running static and modal FEA on differential housing assemblies, comparing solutions to achieve a unit cost reduction of \$100.
- Redesigning and releasing LWL spindle components, four pinions, trunnions, a wheel speed sensor assembly, and mining-truck A-frame cover plates by performing CN revisions and validating designs against APQP standards.

Advanced Blankets and Coolants Laboratory

Urbana, Illinois

Research Assistant

January 2026 - present

- Preparing MoO₃ and He salt-sparging test materials with a graduate researcher.
- Designing an EIS probe to measure noble-metal particle density during helium-sparging experiments.

Hendrickson International

Woodridge, Illinois

Design Intern

October 2023 - August 2024

- Collaborated with the Current Products department to standardize around 120 torque rods with NX, Excel, and Siemens TC.
- Validated BOM files for an e-axle gearbox, updating parts through Teamcenter and NX to verify assembly accuracy.

RWE Design Build

Downers Grove, Illinois

Engineering Intern

February 2023 - May 2023

- Streamlined company information using Procore, communicated with clients regarding master-subcontractor agreements.
- Updated project timelines and formatted salary data in Bidder Editor to facilitate effective project management.

EXTRACURRICULARS

Off-Road Illini Baja SAE

Urbana, Illinois

Treasurer

April 2025 - present

- Co-led a Creo workshop for underclassmen on chain of command, team standards, interface navigation, and assembly modeling.
- Manage a ~50k budget and purchasing for a team of 100-150 people.

Suspension Subteam Member

August 2024 - present

- Designing rear uprights with Top-Op and ANSYS to parametrically shift, reducing possibility of failure
- Designed a jig fixture for upper control arms; created 3D models in Creo and generated DXF files for waterjet fabrication.
- Participated in hands-on testing with a 10-person team to assess the effect of varying spring tensions on CVT performance.

American Society of Mechanical Engineers

Urbana, Illinois

Secretary

May 2025 - 2026

- Managing room reservations, committee calendars, and biweekly meetings; distributing weekly newsletters to 1,900 subscribers.
- Redesigned the club website using HTML and custom CSS.

Hybrid Piano Lead

November 2025 - April 26

- Led a 20-student team to design and build a \$300 digital hybrid silent-system piano for Engineering Open House, modeling and manufacturing muting mechanisms and handling tuning and service. Ask me more about this!

6-Axis Robot Arm

October 2024 - May 2025

- Collaborated with a 10-member mechanical team to design a manufacturable outer shell for a 6-DOF robotic arm, using Fusion 360 CAD models and 3D-printed motor mounts with the system's architecture.